O*NET OnLine

# Database Architects

## 15-1243.00

Bright Outlook | Updated 2026

Design strategies for enterprise databases, data warehouse systems, and multidimensional networks. Set standards for database operations, programming, query processes, and security. Model, design, and construct large relational databases or data warehouses. Create and optimize data models for warehouse infrastructure and workflow. Integrate new systems with existing warehouse structure and refine system performance and functionality.

**Sample of reported job titles:** Data Architect, Data Engineer, Data Officer, Database Analyst, Database Architect, Database Consultant, Database Developer, Database Programmer, Enterprise Architect, Information Architect

**Also see:** Data Warehousing Specialists

Summary | Details | Custom | Easy Read | Veterans | Español

Contents

## Occupation-Specific Information

## Tasks

Save Table: XLSX CSV

10 of 25 displayed

| Importance | Category | Task |
|---|---|---|
| 88 | Core | Develop and document database architectures. |
| 87 | Core | Collaborate with system architects, software architects, design analysts, and others to understand business or industry requirements. |
| 86 | Core | Develop database architectural strategies at the modeling, design and implementation stages to address business or industry requirements. |

| Importance | Category | Task |
|---|---|---|
| 85 | Core | Design databases to support business applications, ensuring system scalability, security, performance, and reliability. |
| 78 | Core | Develop data models for applications, metadata tables, views or related database structures. |
| 78 | Core | Design database applications, such as interfaces, data transfer mechanisms, global temporary tables, data partitions, and function-based indexes to enable efficient access of the generic database structure. |
| 76 | Core | Develop methods for integrating different products so they work properly together, such as customizing commercial databases to fit specific needs. |
| 75 | Core | Create and enforce database development standards. |
| 74 | Core | Document and communicate database schemas, using accepted notations. |
| 74 | Core | Develop data model describing data elements and their use, following procedures and using pen, template or computer software. |

## Occupational Requirements

# Work Activities

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10 of 41 displayed

| Importance | Work Activity |
|---|---|
| 94 | **Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information. |
| 91 | **Analyzing Data or Information** — Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts. |
| 85 | **Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources. |

| Importance | Work Activity |
|---|---|
| 85 | **Processing Information** — Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data. |
| 82 | **Making Decisions and Solving Problems** — Analyzing information and evaluating results to choose the best solution and solve problems. |
| 81 | **Updating and Using Relevant Knowledge** — Keeping up-to-date technically and applying new knowledge to your job. |
| 79 | **Communicating with Supervisors, Peers, or Subordinates** — Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person. |
| 78 | **Identifying Objects, Actions, and Events** — Identifying information by categorizing, estimating, recognizing differences or similarities, and detecting changes in circumstances or events. |
| 73 | **Estimating the Quantifiable Characteristics of Products, Events, or Information** — Estimating sizes, distances, and quantities; or determining time, costs, resources, or materials needed to perform a work activity. |
| 73 | **Evaluating Information to Determine Compliance with Standards** — Using relevant information and individual judgment to determine whether events or processes comply with laws, regulations, or standards. |

# Detailed Work Activities

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10 of 25 displayed

- Create databases to store electronic data.
- Document technical specifications or requirements.
- Collaborate with others to determine design specifications or details.
- Develop procedures for data management.
- Develop database parameters or specifications.
- Develop models of information or communications systems.
- Design computer modeling or simulation programs.
- Develop guidelines for system implementation.
- Develop performance metrics or standards related to information technology.
- Communicate project information to others.

# Work Context

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10 of 57 displayed

**E-Mail** — How frequently does your job require you to use E-mail?

- 95% Every day

**Spend Time Sitting** — How much does this job require sitting?

- 81% Continually or almost continually
- 19% More than half the time

**Telephone Conversations** — How often do you have telephone conversations in this job?

- 48% Every day
- 48% Once a week or more but not every day

**Face-to-Face Discussions with Individuals and Within Teams** — How frequently does your job require face-to-face discussions with individuals and within teams?

- 43% Every day
- 48% Once a week or more but not every day

**Freedom to Make Decisions** — How much decision making freedom, without supervision, does the job offer?

- 29% A lot of freedom
- 62% Some freedom

**Importance of Being Exact or Accurate** — How important is being very exact or highly accurate in performing this job?

- 48% Extremely important
- 33% Very important

**Work With or Contribute to a Work Group or Team** — How important is it to work with or contribute to a work group or team in this job?

- 48% Extremely important
- 29% Very important
- 19% Important

**Indoors, Environmentally Controlled** — How often does this job require working indoors in an environmentally controlled environment (like a warehouse with air conditioning)?

- 65% Every day
- 15% Once a week or more but not every day
- 15% Never

**Determine Tasks, Priorities and Goals** — How much freedom does the worker have in determining the tasks, priorities, or goals of the job?

- 29% A lot of freedom
- 52% Some freedom
- 19% Limited freedom

**Duration of Typical Work Week** — Number of hours typically worked in one week.

- 43% More than 40 hours
- 57% 40 hours

## Experience Requirements

# Job Zone

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| | |
|---|---|
| **Title** | Job Zone Four: Considerable Preparation Needed |
| **Education** | Most of these occupations require a four-year bachelor's degree, but some do not. |
| **Related Experience** | A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of college and work for several years in accounting to be considered qualified. |
| **Job Training** | Employees in these occupations usually need several years of work-related experience, on-the-job training, and/or vocational training. |
| **Job Zone Examples** | Many of these occupations involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database administrators, graphic designers, conservation scientists, art directors, and cost estimators. |
| **SVP Range** | (7.0 to < 8.0) |

# Training & Credentials

**State training** Select a State Go

**Local training** ZIP Code Go

**Certifications** Find certifications nationwide

# Apprenticeship Opportunities

All 2 displayed

Example apprenticeship titles for this occupation:

- Database Architect
- Database Architect (Nof)

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

## Worker Requirements

# Software Skills

**Save Table:** XLSX CSV

10 of 59 displayed

- **Backup or archival software** — Data Recovery Software SQL Server Data Recovery; EMC NetWorker; Oracle Recovery Manager; Veritas NetBackup
- **Data base management system software** — Amazon DynamoDB; Elasticsearch; MongoDB; Oracle PL/SQL
- **Data base user interface and query software** — Apache Hive; Blackboard software; IBM DB2; Transact-SQL
- **Development environment software** — Apache Kafka; Apache Maven; Go; Oracle SQL Developer
- **Enterprise application integration software** — Extensible markup language XML; IBM InfoSphere DataStage; Microsoft SQL Server Integration Services SSIS; SAS Data Integration Studio
- **Enterprise resource planning ERP software** — Ellucian Banner ERP; Microsoft Dynamics; Oracle PeopleSoft; SAP software
- **Metadata management software** — Informatica software; Quest Erwin Data Modeler; SAP PowerDesigner; Talend Data Fabric
- **Object or component oriented development software** — Apache Spark; jQuery; Scala; Swift
- **Operating system software** — Bash; Microsoft Windows Server; Red Hat Enterprise Linux; UNIX Shell
- **Web platform development software** — Django; Google Angular; React; Spring Framework

Hot Technologies are requirements most frequently included across all employer job postings.
See all 112 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 13 In Demand skills for this occupation.

# Essential Skills

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| Importance | Essential Skill |
|---|---|
| 72 | **Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems. |
| 72 | **Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents. |
| 63 | **Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times. |
| 63 | **Speaking** — Talking to others to convey information effectively. |
| 56 | **Active Learning** — Understanding the implications of new information for both current and future problem-solving and decision-making. |
| 53 | **Monitoring** — Monitoring/Assessing performance of yourself, other individuals, or organizations to make improvements or take corrective action. |
| 53 | **Writing** — Communicating effectively in writing as appropriate for the needs of the audience. |
| 50 | **Mathematics** — Using mathematics to solve problems. |
| 47 | **Learning Strategies** — Selecting and using training/instructional methods and procedures appropriate for the situation when learning or teaching new things. |
| 22 | **Science** — Using scientific rules and methods to solve problems. |

# Transferable Skills

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10 of 25 displayed

| Importance | Transferable Skill |
|---|---|
| 72 | **Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions. |
| 72 | **Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one. |
| 69 | **Systems Analysis** — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes. |
| 60 | **Programming** — Writing computer programs for various purposes. |
| 56 | **Systems Evaluation** — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system. |
| 53 | **Operations Analysis** — Analyzing needs and product requirements to create a design. |
| 50 | **Coordination** — Adjusting actions in relation to others' actions. |
| 50 | **Service Orientation** — Actively looking for ways to help people. |
| 50 | **Social Perceptiveness** — Being aware of others' reactions and understanding why they react as they do. |
| 50 | **Technology Design** — Generating or adapting equipment and technology to serve user needs. |

# Knowledge

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10 of 33 displayed

| Importance | Knowledge |
|---|---|
| 83 | **Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming. |
| 74 | **Engineering and Technology** — Knowledge of the practical application of engineering science and technology. This includes applying principles, techniques, procedures, and equipment to the design and production of various goods and services. |

| Importance | Knowledge |
|---|---|
| 68 | **Design** — Knowledge of design techniques, tools, and principles involved in production of precision technical plans, blueprints, drawings, and models. |
| 67 | **Mathematics** — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications. |
| 54 | **English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar. |
| 47 | **Administration and Management** — Knowledge of business and management principles involved in strategic planning, resource allocation, human resources modeling, leadership technique, production methods, and coordination of people and resources. |
| 41 | **Education and Training** — Knowledge of principles and methods for curriculum and training design, teaching and instruction for individuals and groups, and the measurement of training effects. |
| 33 | **Customer and Personal Service** — Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction. |
| 31 | **Law and Government** — Knowledge of laws, legal codes, court procedures, precedents, government regulations, executive orders, agency rules, and the democratic political process. |
| 26 | **Telecommunications** — Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems. |

# Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

| | |
|---|---|
| 76% | Bachelor's degree required |
| 14% | Master's degree required |
| 5% | Post-secondary certificate required |

## Worker Characteristics

# Abilities

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10 of 52 displayed

| Importance | Ability |
|---|---|
| 72 | **Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense. |
| 72 | **Oral Comprehension** — The ability to listen to and understand information and ideas presented through spoken words and sentences. |
| 72 | **Written Comprehension** — The ability to read and understand information and ideas presented in writing. |
| 69 | **Inductive Reasoning** — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events). |
| 69 | **Information Ordering** — The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations). |
| 69 | **Oral Expression** — The ability to communicate information and ideas in speaking so others will understand. |
| 69 | **Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem. |
| 66 | **Near Vision** — The ability to see details at close range (within a few feet of the observer). |
| 66 | **Speech Clarity** — The ability to speak clearly so others can understand you. |
| 60 | **Category Flexibility** — The ability to generate or use different sets of rules for combining or grouping things in different ways. |

# Career Interest Types

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| Occupational Interest | Career Interest Type |
|---|---|
| 85 | **Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources. |
| 77 | **Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service. |
| 28 | **Enterprising** — Work involves managing, negotiating, marketing, or selling, typically in a business setting, or leading or advising people in political and legal situations. Enterprising occupations are often associated with business initiatives, sales, marketing/advertising, finance, management/administration, professional advising, public speaking, politics, or law. |
| 27 | **Realistic** — Work involves designing, building, or repairing of equipment, materials, or structures, engaging in physical activity, or working outdoors. Realistic occupations are often associated with engineering, mechanics and electronics, construction, woodworking, transportation, machine operation, agriculture, animal services, physical or manual labor, athletics, or protective services. |
| 19 | **Artistic** — Work involves creating original visual artwork, performances, written works, food, or music for a variety of media, or applying artistic principles to the design of various objects and materials. Artistic occupations are often associated with visual arts, applied arts and design, performing arts, music, creative writing, media, or culinary art. |
| 15 | **Social** — Work involves helping, teaching, advising, assisting, or providing service to others. Social occupations are often associated with social, health care, personal service, teaching/education, or religious activities. |

## Specific Interest Areas

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2 of 41 displayed

| Occupational Interest | Specific Interest Area |
|---|---|
| 94 | **Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems. |
| 50 | **Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods. |

## Work Styles

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10 of 21 displayed

| Impact | Work Style |
|---|---|
| 100 | **Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work. |
| 84 | **Dependability** — A tendency to be reliable, responsible, and consistent in meeting work-related obligations. |
| 79 | **Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects. |
| 70 | **Achievement Orientation** — A tendency to establish and maintain personally challenging work-related goals, set high work-related standards, and exert high effort toward meeting those goals and standards. |
| 69 | **Cautiousness** — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work. |
| 68 | **Innovation** — A tendency to be inventive, to be imaginative, and to adopt new perspectives on ways to accomplish work. |
| 63 | **Integrity** — A tendency to be honest and ethical at work. |
| 57 | **Adaptability** — A tendency to be open to and comfortable with change, new experiences, or ideas at work. |
| 55 | **Perseverance** — A tendency to exhibit determination and resolve to perform or complete tasks in the face of difficult circumstances or obstacles at work. |
| 53 | **Tolerance for Ambiguity** — A tendency to be comfortable with ambiguity and uncertainty at work. |

## Workforce Characteristics

# Wages & Employment Trends

**Median wages (2025)** $67.07 hourly, $139,500 annual

**State wages** Select a State Go

**Local wages** ZIP Code Go

**Employment (2025)** 69,500 employees

**Projected growth (2025-2035)** ■■■■ Much faster than average (7% or higher)

**Projected job openings (2025-2035)** 3,900

**State trends** Select a State Go

**Top industries (2025)**
Professional, Scientific, and Technical Services (35% employed in this sector)
Finance and Insurance (15%)
Information (15%)

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

# Job Openings on the Web

**State job openings** Select a State Go

**Local job openings** ZIP Code Go

## More Information

# Related Occupations

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10 of 20 displayed

- 15-1299.07 Blockchain Engineers Bright Outlook
- 15-1241.00 Computer Network Architects
- 15-1211.00 Computer Systems Analysts

- 15-1299.08 Computer Systems Engineers/Architects
- 15-2051.00 Data Scientists
- 15-1243.01 Data Warehousing Specialists
- 15-1242.00 Database Administrators
- 15-1244.00 Network and Computer Systems Administrators
- 15-1252.00 Software Developers
- 15-1253.00 Software Quality Assurance Analysts and Testers

# Professional Associations

**Disclaimer:** Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

## National Associations

- Higher Education Data Warehousing ALLY
- Association for Computing Machinery
- Computing Research Association
- IEEE Computer Society
- National Center for Women and Information Technology

## Accreditation, Certification, & Unions

- Institute for Certification of Computing Professionals ALLY
- CompTIA
- DAMA International